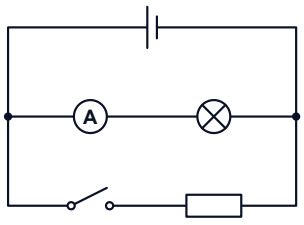
DO NOW · AO1 REVISION

Series and parallel circuits

Electricity
Paper 1 · Combined and Separate
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Name the two ways components can be joined in a circuit.**	In series and in parallel	☐
2 ★	**Describe the current through components in series.**	It is the same through each component.	☐
3	**Describe the potential difference across components in series.**	The total potential difference is shared between them.	☐
4	**Describe the total resistance of components connected in series.**	It is the sum of the individual resistances.	☐
5	**Describe the current through components in parallel.**	The total current is the sum of the currents in the branches.	☐
6	**The lamp in this circuit is lit. Describe what happens to the ammeter reading when the switch is closed.** 	It does not change. The lamp is on a separate branch, so the potential difference across it stays the same.	☐
7 ★	**Describe the potential difference across components in parallel.**	It is the same across each component.	☐
8 ★	**Describe the total resistance of two resistors connected in parallel.**	It is less than the resistance of the smaller resistor.	☐
9	**Explain why adding resistors in series increases the total resistance.**	The potential difference is shared between more resistors, so the current is smaller.	☐
10 ★	**Explain why adding resistors in parallel decreases the total resistance.**	Each new branch gives the charge another path, so the total current increases.	☐

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	QUESTION	ANSWER	✓
11	**Lamps are connected in series and one breaks. State what happens to the others.**	They all go out, because the circuit is broken.	☐
12	**Give two advantages of connecting lamps in parallel rather than in series.**	Each lamp can be switched separately, and if one breaks the others stay lit.	☐

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